

SMART INDIA HACKATHON 2026



- **Problem Statement ID:** 26124
- **Problem Statement Title:** AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:** 135094
- **Team Name:** Coding Wizards



Every bus journey becomes an observation opportunity — UrbanSense determines what can be trusted, what remains uncertain, and which bus should observe next.

Real-world issue:

Urban roads face **potholes, damaged infrastructure, waterlogging, traffic congestion, accidents, hit-and-run, rash driving, and vulnerable-pedestrian risks**, while authorities rely mainly on **fixed CCTV, manual inspections, and public complaints**. Bus cameras are largely underused, causing **delayed hazard/incident detection, incomplete coverage, inefficient maintenance, and increased road-safety risks**.

Core Problem: Cities lack an intelligent system that turns continuously moving buses into **real-time, reliable road, traffic and safety intelligence** for **timely action and prevention**.



INNOVATION & UNIQUENESS — BEYOND DETECTION

01 — OBSERVATION-AWARE EVIDENCE:

Trust a **detection** only when the **road** was actually **observable**.

02 — EVIDENCE-DRIVEN FLEET FUSION:

Combine observations from **multiple buses** using **quality, confidence, time and spatial-temporal consistency**.

03 — EVIDENCE DEBT → NEXT-BEST BUS

When evidence is **insufficient**, select the **next valuable bus** observation to **reduce uncertainty**.

04 — LIFECYCLE-AWARE ROADTWIN

Track every road issue from **Detect → Confirm → Repair → Verify → Reappear**.

05 — CLOSED-LOOP VERIFICATION

Use subsequent fleet observations to **independently verify repairs** and **detect recurring defects**.

DETECTION → TRUSTED EVIDENCE → ACTION → VERIFICATION

BUS FLEET SENSING



EVIDENCE FUSION



LIVE ROADTWIN



AUTHORITY ACTION



Proposed Solution:

- **Edge AI:** Equip buses with **cameras, GPS/GNSS and IMU** to continuously capture road, traffic and safety conditions while moving.
- **Real-Time Urban Perception:** On-board AI detects **road defects, missing infrastructure, waterlogging, vehicles, traffic density, bottlenecks, vulnerable pedestrians and safety hazards**, with tracking and counting.
- **Evidence Engine:** Convert observations into **GPS-tagged, time-stamped, confidence- and quality-scored evidence**, while using **spatial-temporal fusion and multi-bus corroboration** to reduce false and duplicate events.
- **Multi-Bus Incident Reconstruction:** Partial observations from different buses—such as **vehicle appearance, trajectory and ANPR/OCR** are **spatio-temporally** combined to reconstruct and corroborate **potential hit-and-run or rash-driving incidents**.
- **RoadTwin & Maintenance Intelligence:** Maintain a continuously updated RoadTwin for each road segment, tracking **condition, history, maintenance priority and lifecycle** — **Detected → Confirmed → Repaired → Verified → Reappeared**.
- **Central Urban Intelligence:** Aggregate fleet observations into **GIS road-condition maps, congestion heatmaps, infrastructure-deficiency maps, traffic/OD analytics, route-delay insights and incident reports**, enabling evidence-based authority action.
- **Priority-Aware Communication:** **Critical safety and incident evidence** is transmitted immediately over **4G/5G**, while non-urgent observations are **buffered and bulk-synced through depot Wi-Fi**; duplicate or previously observed events are filtered to reduce unnecessary transmission.
- **Proactive Verification:** When evidence is insufficient, the system identifies the **next valuable bus observation** opportunity for confirmation rather than relying on blind repeated sensing.
- **GenAI Authority Copilot:** Authorities can query **RoadTwin, GIS, incidents, traffic, maintenance and approved documents** in natural language; the Copilot retrieves **RBAC-authorized evidence** and generates **grounded, explainable decision support**.

TECH STACK

EDGE SENSING & AI

- **Cameras + OpenCV** — Multi-camera capture & frame preprocessing
- **YOLO + Segmentation** — Road defects, vehicles, pedestrians & hazards
- **Tracking** — Vehicle identity, trajectory & motion continuity
- **OCR / ANPR** — Number-plate localization & text extraction
- **GPS/GNSS + IMU** — Location, heading, motion & sensor context
- **PyTorch → ONNX / TensorRT**—Model training & edge inference optimization

EDGE EVIDENCE & COMMUNICATION

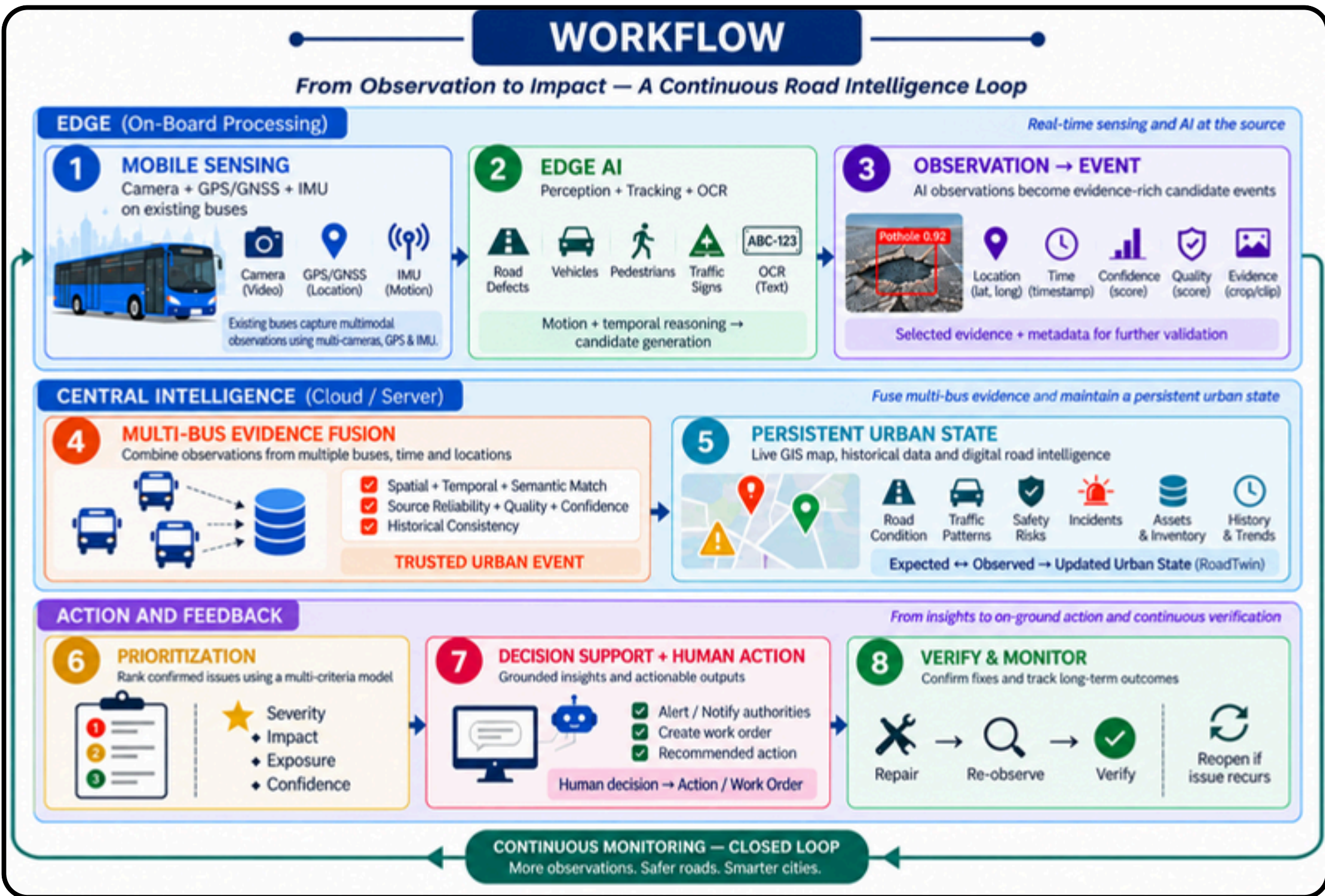
- **Event Engine** — Converts AI outputs into structured observations/events
- **Quality + Confidence Engine** — Visibility, blur, occlusion, GPS & detection reliability
- **Evidence Manager** — Selects frames/clips + metadata for transmission
- **Priority Manager (P0–P4)** — Determines urgency & bandwidth priority
- **Local Queue** — Store-and-forward during connectivity loss
- **MQTT / HTTPS** — Secure event transmission

CENTRAL INTELLIGENCE

- **FastAPI** — Secure APIs for fleet, events, incidents & RoadTwin
- **PostgreSQL + PostGIS** — Urban data storage + spatial/road queries
- **Fusion Engine** — Deduplication + spatial-temporal correlation + multi-bus corroboration
- **RoadTwin Engine** — Persistent current + historical road state & lifecycle
- **Analytics Engine**—Vehicle counts,density, congestion,OD proxy&route delays
- **Maintenance + Alert Engine** — Priority, alerts, department routing & lifecycle actions

GIS, AUTHORITY & GENAI

- **React + TypeScript** — Authority dashboards & operational interfaces
- **MapLibre / Leaflet** — RoadTwin, routes, events & GIS heatmaps
- **LLM + RAG / Tool Retrieval** — Natural-language, evidence-grounded authority Copilot
- **RBAC + Authentication** — Role-scoped access to sensitive evidence
- **Docker** — Reproducible laptop prototype → future edge deployment



SIH Prototype: Laptop + Recorded Video + Simulated GNSS/IMU
→ **Same Edge Agent & Event API** ←
Future Deployment: RTSP / MIPI Cameras + GNSS/IMU + Edge-AI Hardware

TECHNICAL FEASIBILITY:

Can we build it?

- **Edge AI:** YOLO/segmentation, tracking and OCR/ANPR enable onboard perception.
- **Sensor Fusion:** Camera + GNSS/IMU provide visual, location and motion context.
- **Edge Optimization:** PyTorch models can be optimized through ONNX/TensorRT for edge inference.
- **Central Intelligence:** FastAPI + PostgreSQL/PostGIS support event ingestion, spatial queries and RoadTwin.
- **Evidence Pipeline:** Quality/confidence scoring → evidence packets → spatial-temporal fusion → trusted urban events.
- **Prototype Continuity:** Recorded video + simulated GNSS/IMU use the same Edge Agent & Event API planned for live deployment.

Established AI + geospatial + edge technologies → integrated into one deployable pipeline.

DEPLOYMENT & DATA FEASIBILITY:

Can buses generate reliable evidence?

- **Existing Fleet:** Leverages public buses already traversing city roads; no dedicated sensing-vehicle network required.
- **Multimodal Inputs:** Existing/retrofit cameras + GNSS/GPS + IMU provide complementary observations.
- **Quality-Aware Evidence:** Visibility, blur, occlusion, location quality and detection confidence determine evidence reliability.
- **Bandwidth Efficient:** Edge filtering converts continuous video into compact, prioritized evidence/events.
- **Connectivity Resilience:** Local store-and-forward queue preserves observations during network loss.
- **Incremental Retrofit:** Begin with selected buses/routes and progressively expand across the fleet.

Reuse mobility infrastructure → process locally → transmit only useful evidence.

OPERATIONAL VIABILITY:

Can authorities act on it continuously?

- **Actionable Outputs:** GIS road-condition maps, congestion insights, infrastructure deficiencies and incident reports.
- **Evidence-Based Prioritization:** Issues ranked using severity × impact × exposure × confidence.
- **Authority Workflow:** Detection → confirmation → priority → action/work order → repair → verification.
- **Closed-Loop Monitoring:** Subsequent buses independently re-observe repaired segments and detect reappearance.
- **Human-in-the-Loop:** AI provides evidence, prioritization and grounded decision support; authorized personnel take action.
- **GenAI Copilot:** Natural-language access to authorized RoadTwin, GIS, incidents and maintenance evidence.

UrbanSense doesn't stop at detection — it connects evidence to action and verification.



REAL-WORLD RISKS → ARCHITECTURAL RESPONSE

01 — OBSERVATION RELIABILITY

RISK: Visibility, blur & occlusion can make detections unreliable.

RESPONSE:

Quality + observability scoring
→ Observation - Qualification Gate
→ Poor passes remain **INCONCLUSIVE**, not false negatives.

02 — LOCATION RELIABILITY

RISK: GNSS drift or temporary signal loss can misplace events.

RESPONSE:

GNSS + IMU + heading + map matching
→ Location confidence + multi-pass consistency.

03 — EVIDENCE INTEGRITY

RISK: Observations can be false, duplicated or insufficient.

RESPONSE:

Spatial-temporal fusion + multi-bus corroboration
→ Explicit **Insufficient Evidence** state
→ **Next-Best Bus**.

04 — COMMUNICATION RESILIENCE

RISK: Connectivity issues can disrupt evidence delivery.

RESPONSE:

Edge filtering + priority routing
→ Durable local queue
→ Retry / backoff + depot synchronization.

05 — DATA SECURITY

RISK: ANPR and incident evidence require controlled access.

RESPONSE:

RBAC + TLS-secured HTTPS + MQTT
→ Role-scoped access + evidence provenance.

INCREMENTAL DEPLOYMENT: Prototype → Pilot Routes → Fleet → City → Multi-City

RETROFIT-FRIENDLY · BANDWIDTH-AWARE · MODULAR · HUMAN-IN-THE-LOOP

FROM REACTIVE MONITORING → CONTINUOUS URBAN INTELLIGENCE

OBSERVE

Road • Traffic • Safety

EVIDENCE

Quality • Confidence • Fleet Corroboration

ACT

Prioritize • Alert • Work Order

VERIFY

Repair • Re-observe • Reappearance



BUS OPERATORS

BENEFITS:

- **Detects** severe road hazards and waterlogging early along active routes.
- **Optimizes bandwidth** through priority-aware communication: urgent evidence first, non-urgent data batched for later synchronization.

IMPACT:

SMARTER FLEET OPERATIONS

From dedicated inspection vehicles → To continuous, fleet-generated road intelligence.



URBAN AUTHORITIES

BENEFITS

- **Automatically ranks** issues using **Severity × Traffic Impact × Exposure × Confidence**.
- **Empowers** decision-makers with a **grounded GenAI Copilot** for natural-language queries across authorized GIS, RoadTwin and operational data.

IMPACT:

EVIDENCE-DRIVEN ACTION

From reactive, complaint-led responses → To prioritized, evidence-backed decisions.

ROAD / MAINTENANCE
DEPARTMENTS

BENEFITS

- **Tracks the complete lifecycle** of every issue within RoadTwin: **Detected → Confirmed → Repair → Verified → Reappeared**.
- **Flags unresolved or recurring defects** through repeated post-repair observations.

IMPACT:

CLOSED-LOOP MAINTENANCE

From one-time manual inspections → To continuous, evidence-backed repair verification.



TRAFFIC / POLICE

BENEFITS

- **Correlates cross-bus ANPR/OCR** observations, aligning vehicle attributes, trajectory and time/location context.
- **Generates compact, traceable evidence packages** instead of continuously transferring raw video.

IMPACT:

CORROBORATED INCIDENT EVIDENCE

From fragmented, isolated observations → To corroborated, investigation-ready evidence.



CITIZENS & COMMUTERS

BENEFITS

- **Detects** road hazards and **vulnerable-road-user risks** earlier, supporting faster authority attention.
- **Provides timely congestion, bottleneck and route-delay intelligence** to support more predictable journeys.

IMPACT:

SAFER, MORE PREDICTABLE TRAVEL

From unexpected delays and hazards → To earlier awareness and more informed journeys.



SECONDARY ENVIRONMENTAL BENEFITS

Less unnecessary data transmission • Fewer dedicated inspection journeys where bus observations can serve the monitoring need • Data-informed traffic management.

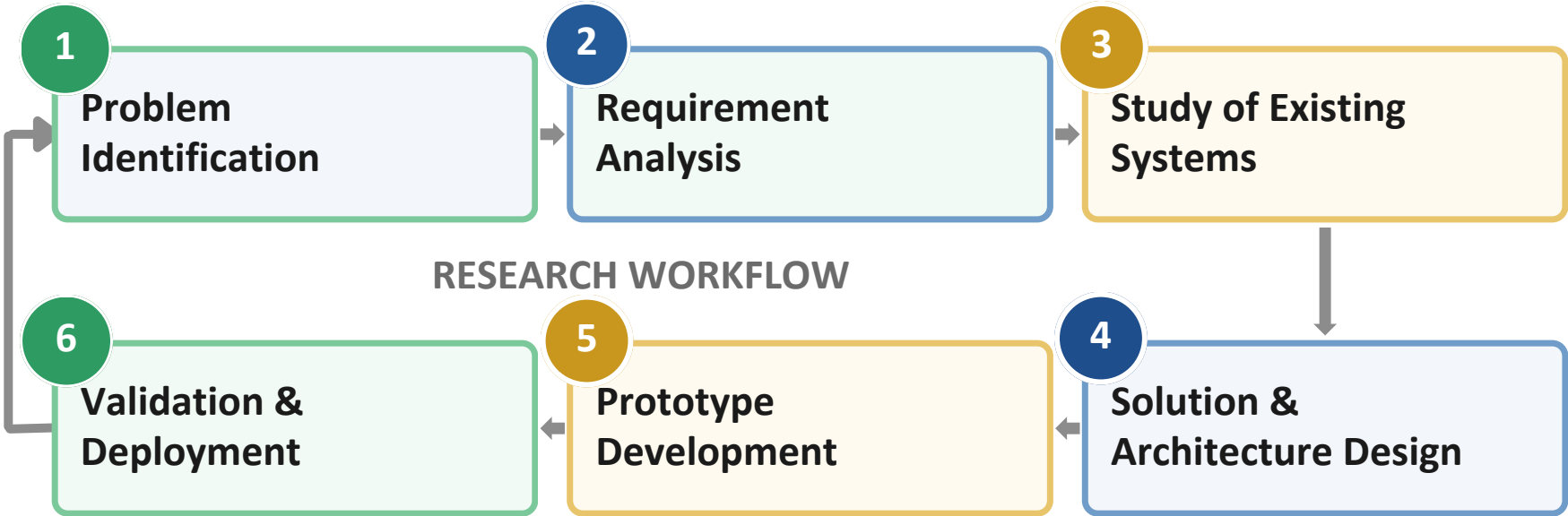
◆ REFERENCES

Research & Best Practices

- [BusEdge](#) — Edge Video Analytics for Transit Buses (*Canbo Ye, CMU, 2021*)
- [Road Condition Monitoring via Bus Trajectories](#) (*Multimodal Transportation, 2022*)
- [Reliability-Aware Crowdsensing for Pothole Profiling](#) (*Zhong et al., ACM IMWUT, 2020*)
- [Truth Discovery in Crowdsourced Spatial Events](#) (*Robin Wentao Ouyang, IEEE TKDE*)
- [Edge Intelligence + Digital Twin for Pothole Detection](#) (*IEEE IoT, 2025*)
- [RDD2022](#) — Multi-National Road Damage Dataset

◆ EXISTING SYSTEMS

- [iRASTE](#) — IIIT-H, Intel & Nagpur Municipal Corp. — AI dashcams on 350+ city buses for driver alerts & road-safety mapping.
- [VIOLA \(IIIT-H / INAI\)](#) — Dashcams on police vehicles auto-detect helmet/lane violations + ANPR for e-challans; extending to potholes.
- [Netradyne Driver-i](#) — Edge-AI fleet dashcam (India & global) scoring driver behaviour in real time — not focused on road-infra defects.
- [City ITMS / ANPR networks](#) — Fixed junction cameras across Indian cities auto-issue challans — static coverage, no mobile defect sensing.
- [RoadBotics](#) — Smartphone-on-windshield AI rates pavement condition city-wide (US/global) via one-time survey drives, not a live feed.



◆ DATASETS & BENCHMARKS

Module / Requirement	Primary Data Source	Type
Indian vehicles & road scenes	IDD (India Driving Dataset)	Public
Vehicle detection & tracking	IDD + BDD100K (MOT)	Public
Pedestrian detection	IDD + BDD100K	Public
Road segmentation	IDD	Public
Potholes & road damage	Road Damage Dataset + custom capture	Hybrid
Traffic signs	Public sign dataset + custom India data	Hybrid
Waterlogging & ANPR	Public plate/image data + custom footage	Hybrid
Fleet GPS, incidents & GIS	Own recorded GPS, video & route/event logs	Custom

◆ RESEARCH / SYSTEM LANDSCAPE

Capability	UrbanSense	iRASTE	Netradyne	ITMS/ANPR
Mobile bus-fleet edge-AI sensing	✓	✓	✓	✗
Pothole / road-infra defect detection	✓	Partial	✗	✗
ANPR & incident / hit-and-run tracking	✓	✗	✗	✓
Persistent defect lifecycle (RoadTwin)	✓	✗	✗	✗
Traffic density / OD mobility analytics	✓	Partial	✗	Partial
GIS dashboard for city authorities	✓	Partial	✗	✓
Low-cost pilot on existing fleet	✓	✗	✗	✗